Advanced Natural Language Processing

AI Agents – Introduction

Sher Muhammad Daudpota

Sukkur IBA University

AI Agents

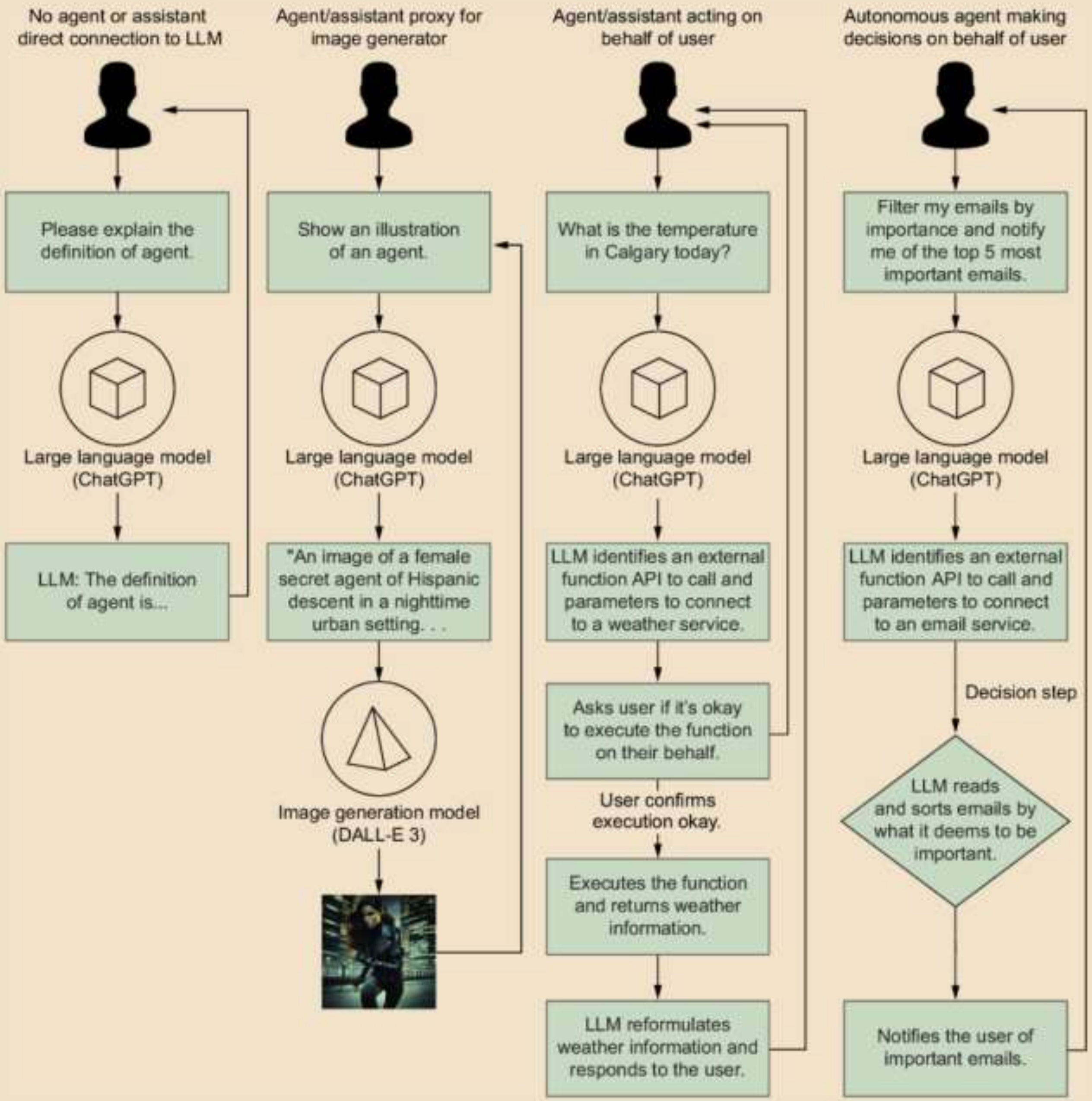


Figure 1.1 The differences between the LLM interactions from direct action compared to using proxy agents, agents, and autonomous agents

AI Agents

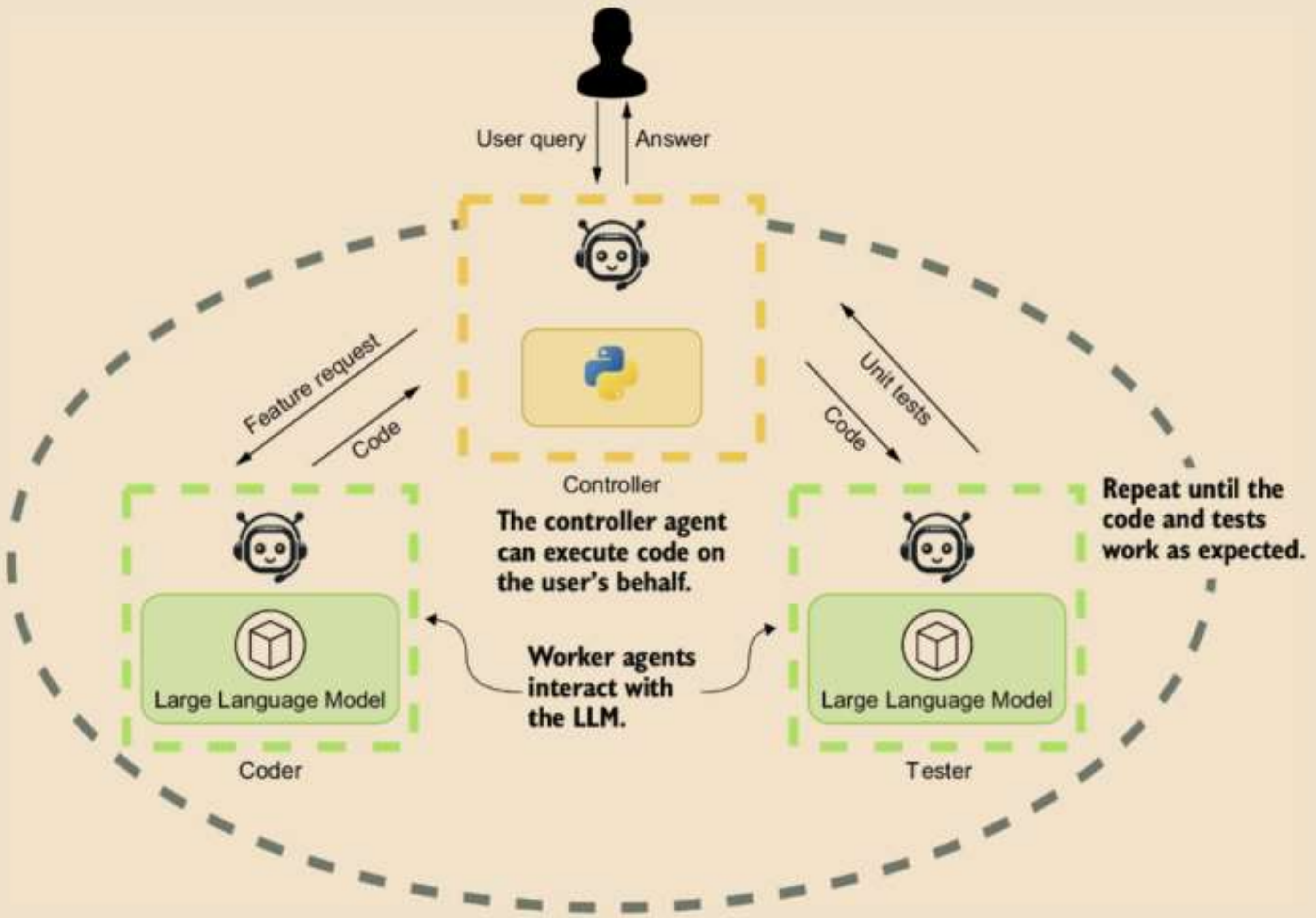


Figure 1.2 In this example of a multi-agent system, the controller or agent proxy communicates directly with the user. Two agents—a coder and a tester—work in the background to create code and write unit tests to test the code.

Component of an AI Agent

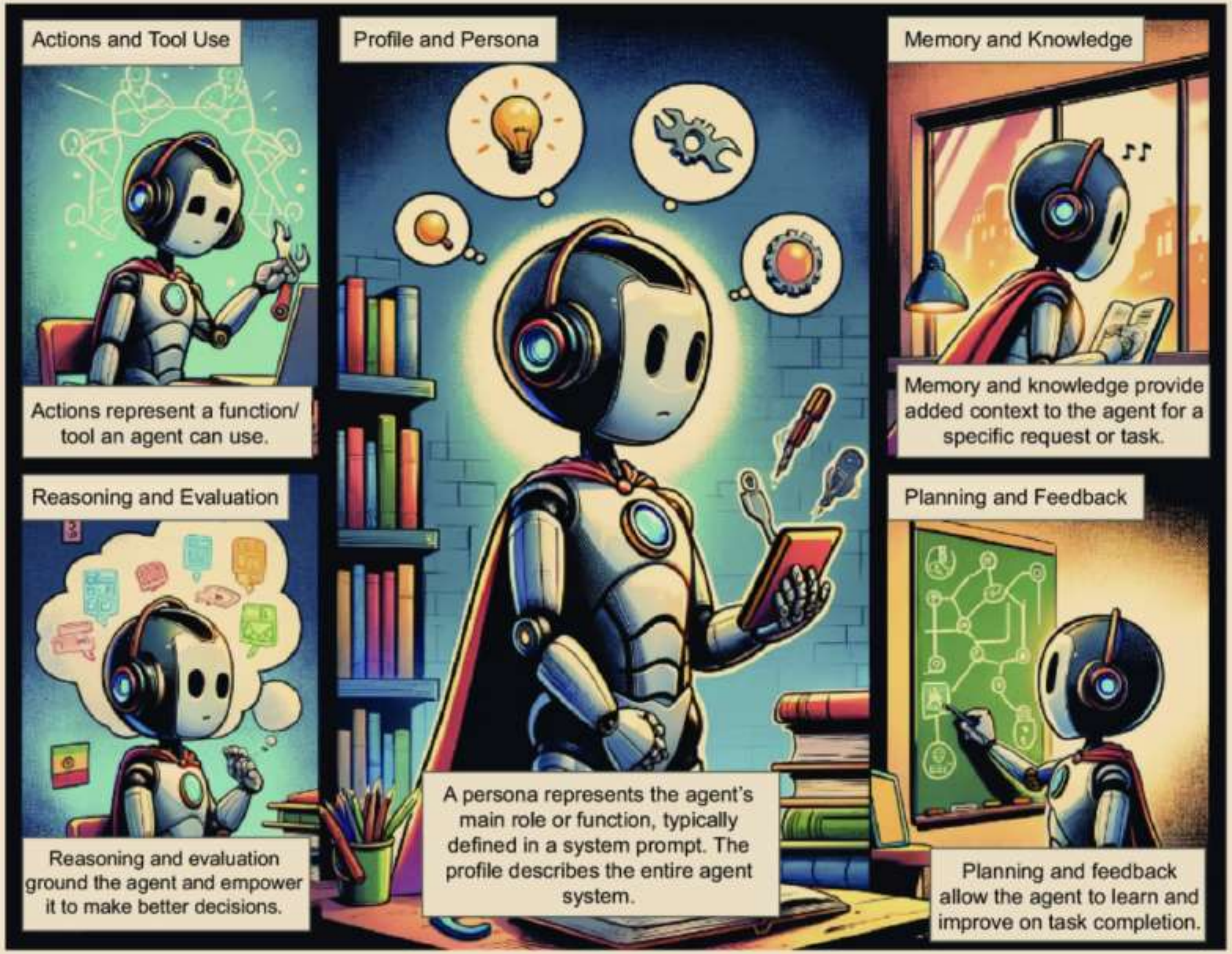


Figure 1.3 The five main components of a single-agent system (image generated through DALL-E 3)

Agent Profile

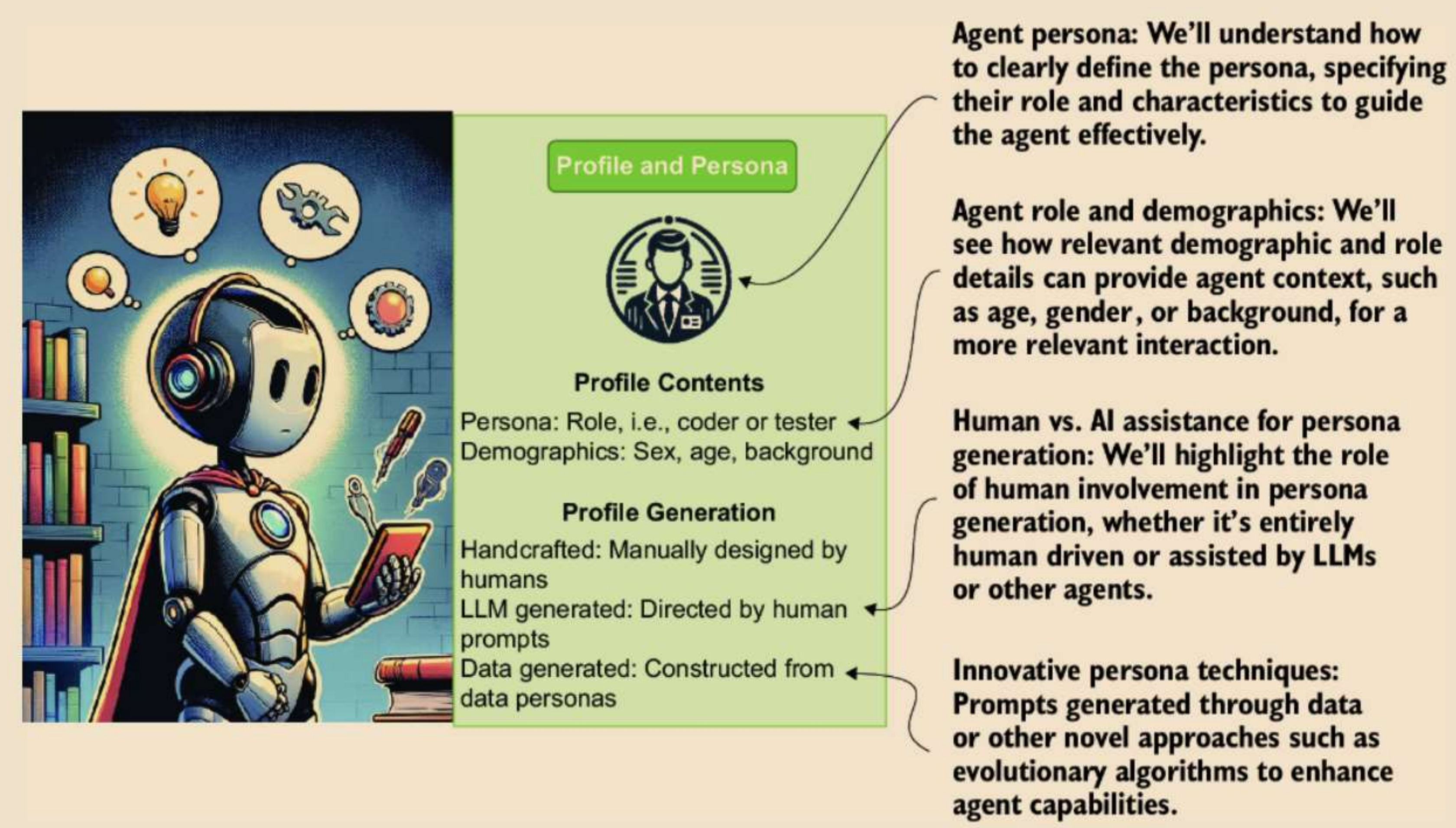
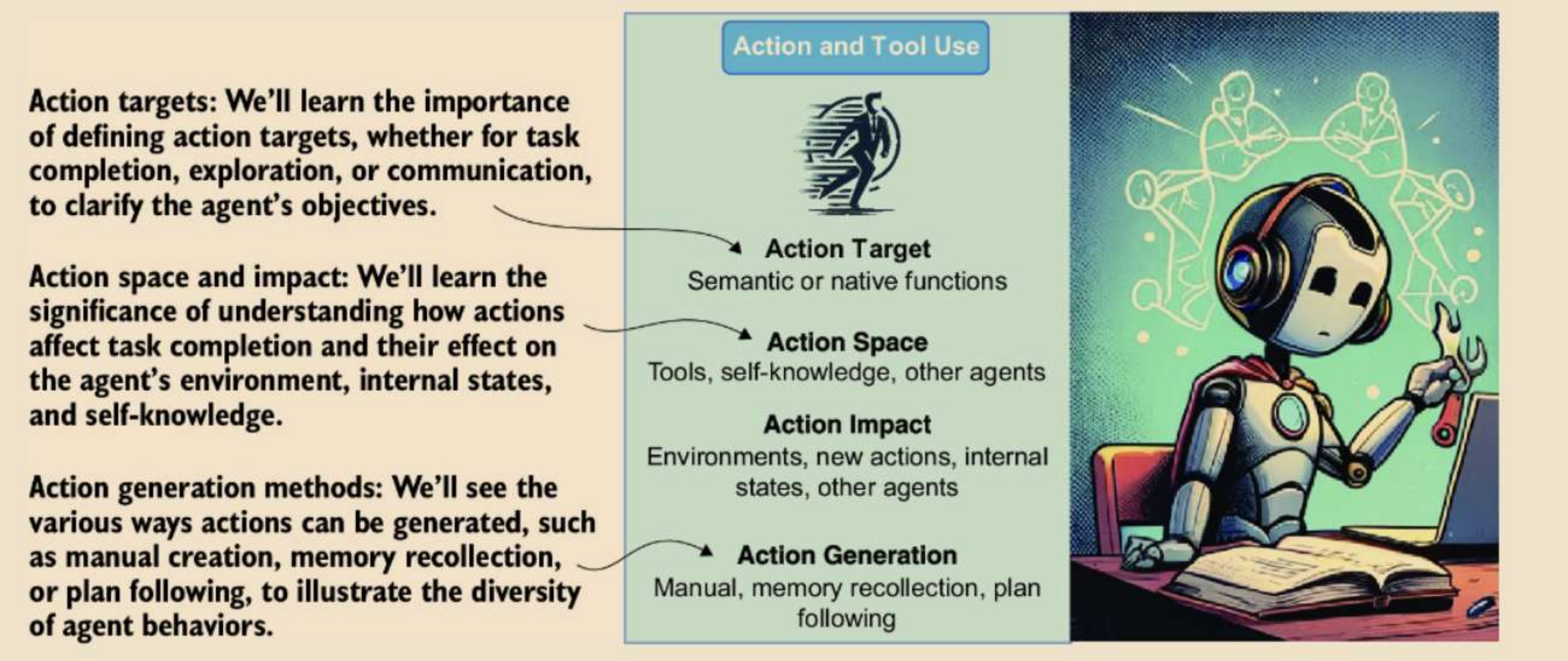
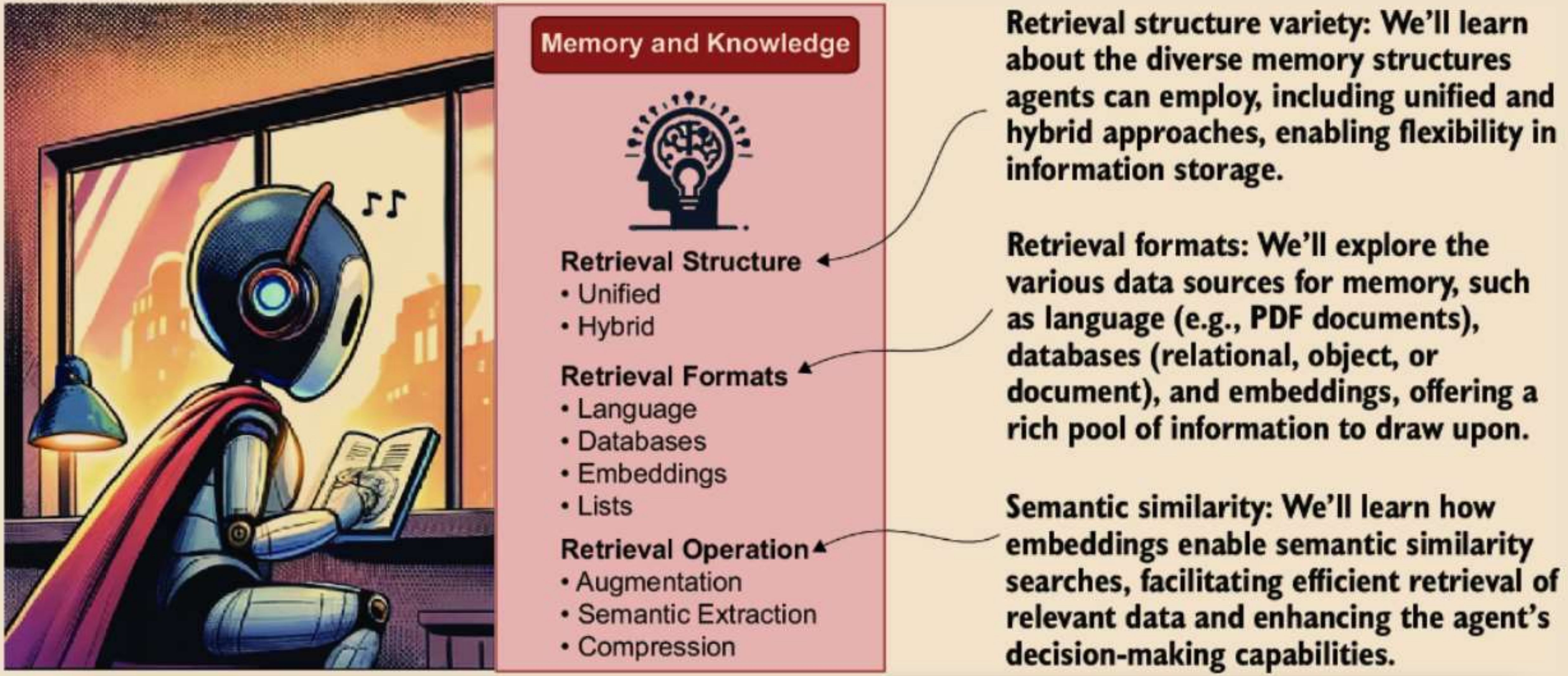


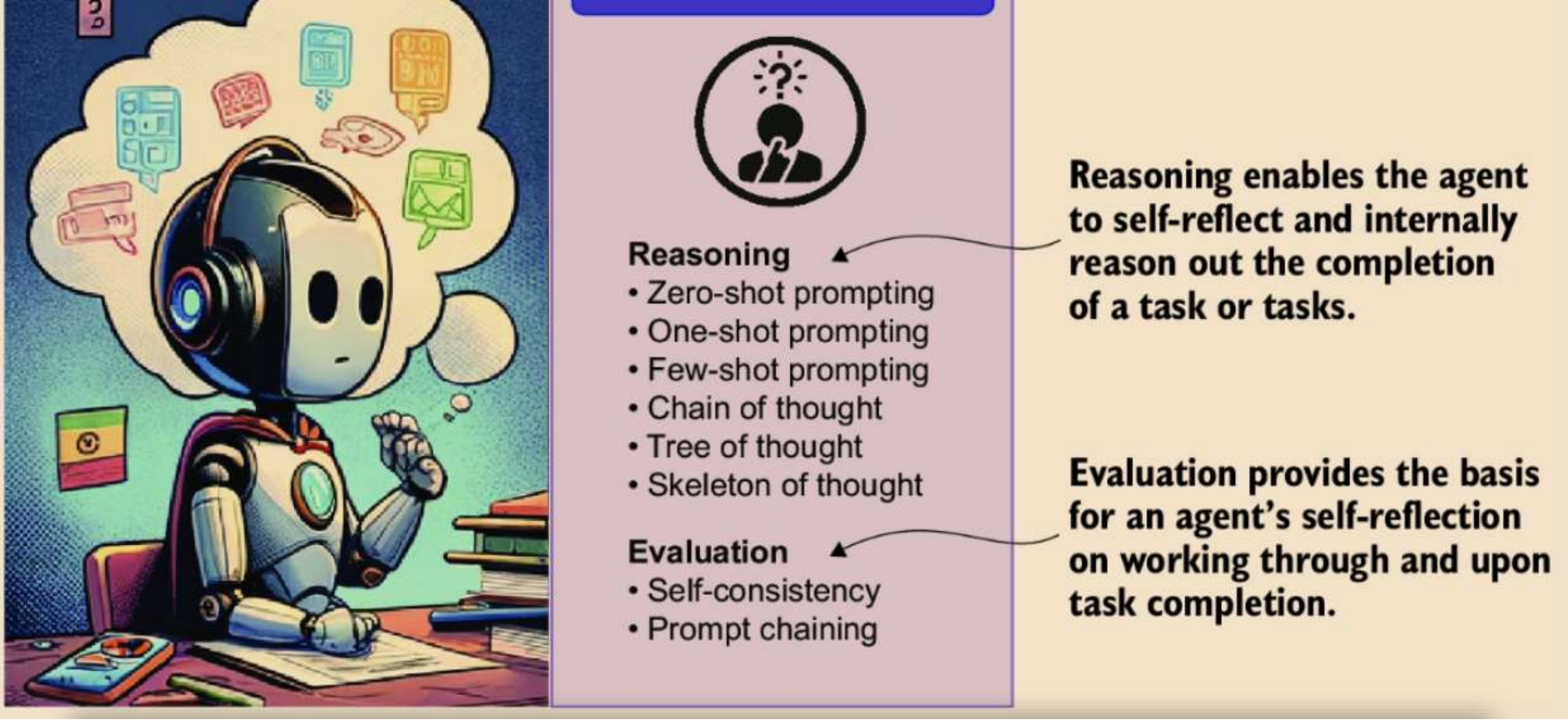
Figure 1.4 An in-depth look at how we'll explore creating agent profiles



Agent Actions & Tools



Agent Memory & Knowledge



Agent Reasoning & Evaluation

Agent Planning & Feedback

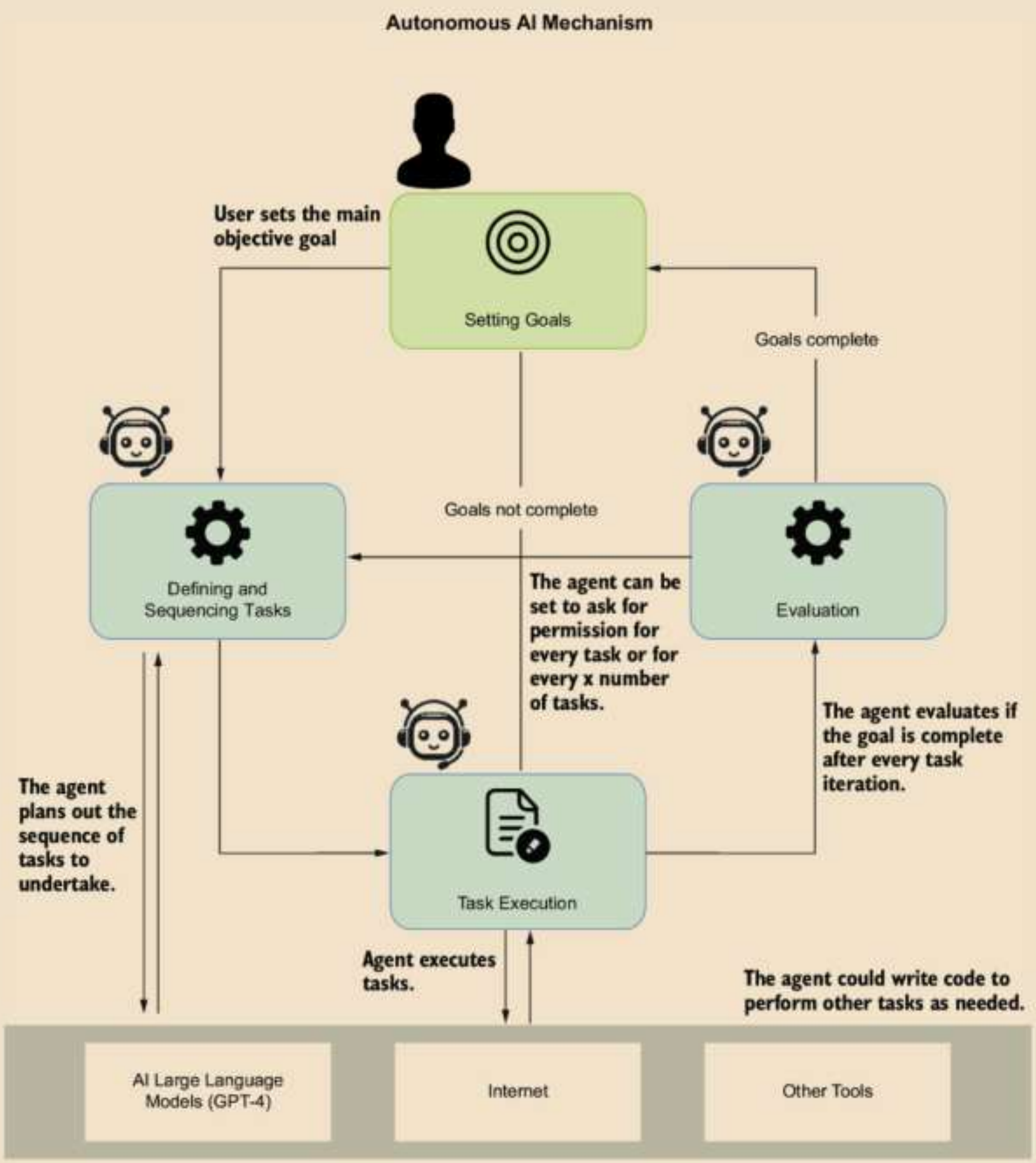


Figure 1.9 The original design of the AutoGPT agent system

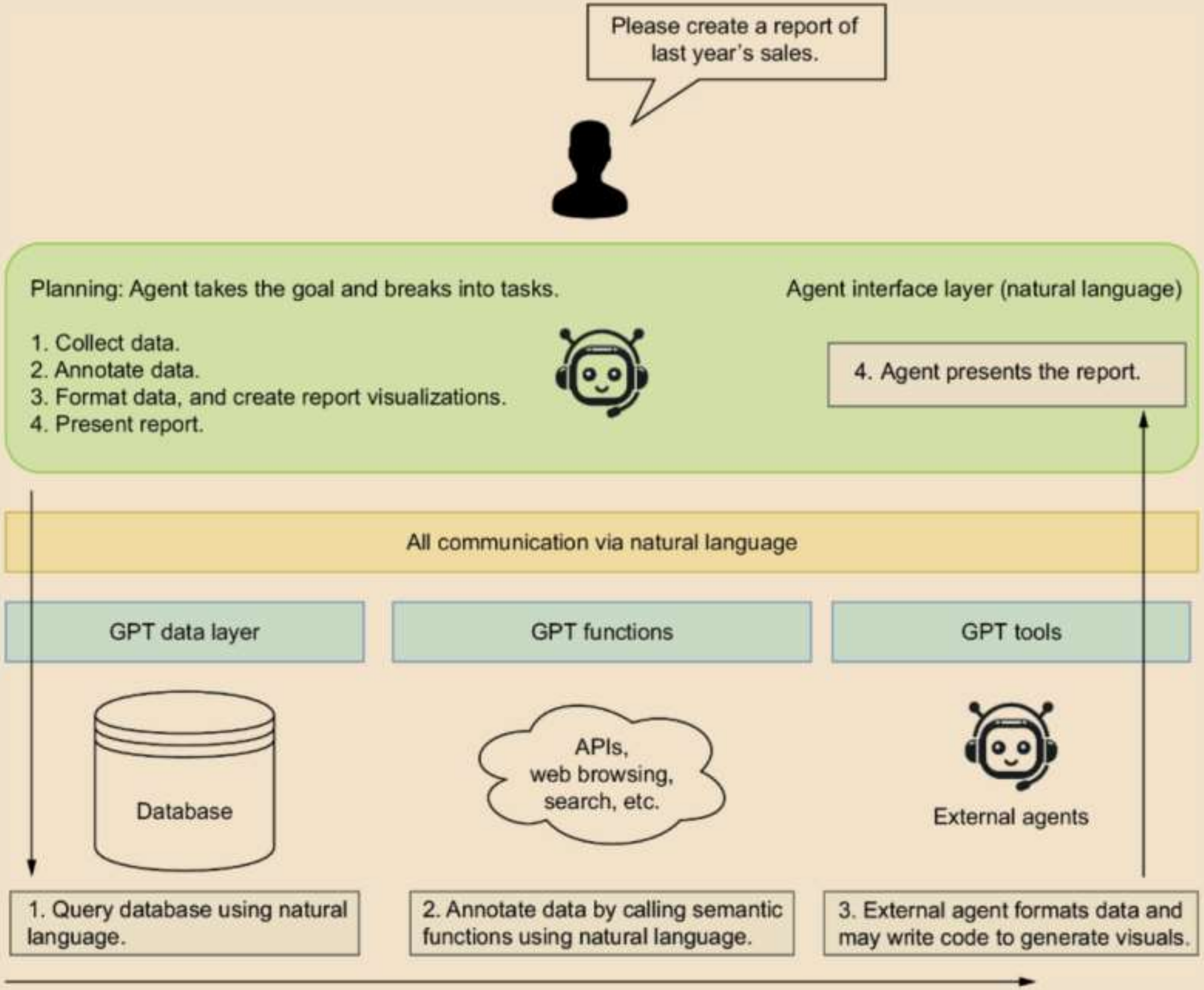


Figure 1.10 A vision of how agents will interact with software systems

- An agent is an entity that acts or exerts power, produces an effect, or serves as a means for achieving a result. An agent automates interaction with a large language model (LLM) in AI.
- An assistant is synonymous with an agent. Both terms encompass tools such as OpenAI's GPT Assistants.
- Autonomous agents can make independent decisions, and their distinction from non-autonomous agents is crucial.

- The four main types of LLM interactions include direct user interaction, agent/ assistant proxy, agent/assistant, and autonomous agent.
- Multi-agent systems involve agent profiles working together, often controlled by a proxy, to accomplish complex tasks.
- The main components of an agent include the profile/persona, actions, knowledge/memory, reasoning/evaluation, and planning/feedback.
- Agent profiles and personas guide an agent's tasks, responses, and other nuances, often including background and demographics.

- Actions and tools for agents can be manually generated, recalled from memory, or follow predefined plans.
- Agents use knowledge and memory structures to optimize context and minimize token usage via various formats, from documents to embeddings.
- Reasoning and evaluation systems enable agents to think through problems and assess solutions using prompting patterns such as zero-shot, one-shot, and few-shot.
- Planning/feedback components organize tasks to achieve goals using single-path or multipath reasoning and integrating environmental and human feedback.
- The rise of AI agents has introduced a new software development paradigm, shifting from traditional to natural language–based AI interfaces.

- Understanding the progression and interaction of these tools helps develop agent systems, whether single, multiple, or autonomous.